

Beach Sand Mineral Image Classification

Reproduction and evaluation of “D-ResNet: deep residual neural network for exploration, identification and classification of beach sand minerals” (Theerthagiri & Prasannavenkatesan, 2022)

Dataset → EDA → preprocessing → four-architecture training → 24-trial hyperparameter study

Item	Detail
Dataset	Kaggle: prasannavenkatesan/beach-sand-mineral-bsm (367 images, 6 classes)
Models	VGG16, ResNet18, ResNet34, D_ResNet (torchvision ResNet50 approximation)
Parameters	134.29 M / 11.18 M / 21.29 M / 23.52 M respectively
Baseline training	60 epochs per model (240 total), batch 32, Adam 1e-4, seed 42
Tuning study	20 epochs per trial × 24 trials (480 epochs), 2 configs × 3 seeds × 4 models
Hardware	NVIDIA Tesla T4 (Colab / Kaggle), CUDA, mixed precision
Best result	0.8036 test accuracy (ResNet18, baseline)
Tuning outcome	Test accuracy improved on all 4 models; gain inside seed noise

We reproduce the experimental setting of a published BSM classification study and then measure how much of the reported result is attributable to the configuration rather than to seed variation. All Python lives in five executed notebooks; every figure below is read from the committed result CSVs.

1. Dataset and exploratory data analysis

The dataset is beach sand mineral (BSM) microscope imagery from Kaggle (prasannavenkatesant/beach-sand-mineral-bsm, public), mirrored at Mendeley Data (10.17632/9kyrz3shk8). It holds microscope photographs of six heavy-mineral grains: **Ilmenite** (FeTiO_3), **Rutile** (TiO_2), **Zircon** (ZrSiO_4), **Garnet** (dodecahedral silicate), **Monazite** ($(\text{Ce,La,Nd,Th})\text{PO}_4$) and **Sillimanite** (Al_2SiO_5).

1.1 Class distribution and the paper discrepancy

Class	Actual on disk	Paper-reported	Difference
Ilmenite	62	60	+2
Rutile	66	62	+4
Zircon	59	60	-1
Garnet	60	66	-6
Monazite	60	60	+0
Sillimanite	60	59	+1
TOTAL	367	367	+0
TOTAL	367	367	0

Table 1.1 — Class distribution. The paper’s text states 357 images, but its own six class counts sum to 367. We report on-disk truth and never force a total to match the published figure.

Counts are near-balanced (59–66 per class), so accuracy is a reasonable primary metric. We nevertheless report macro-averaged precision, recall and F1 throughout, because macro-averaging weights every class equally and cannot be inflated by a majority class.

1.2 EDA performed

EDA step	Finding
Full file inventory	367 files, all readable, 0 corrupt or undecodable
Class distribution	Near-balanced, 59–66 per class (Table 1.1)
Image dimensions	Variable; the minimum side limits the resize step
Visual sample grid	Background composition differs systematically by class
Duplicate detection	0 exact duplicates; 367 unique content hashes
Data wrangling	0 images removed — nothing was lost

Table 1.2 — EDA steps and outcomes, executed in notebook 01.

The most consequential EDA finding is methodological. Visual inspection showed that background composition correlates with class — a Garnet field is visibly pink, an Ilmenite field grey-black. A CNN can therefore reach high accuracy by learning background colour rather than grain morphology. A prior audit of this same dataset quantified the association at **Cramer’s $V \neq 0.442$** . We therefore treat every accuracy figure in this report as an **upper bound** on genuine mineral identification rather than presenting a number we know is inflated.

2. Data preprocessing

2.1 Wrangling and augmentation

Validation found no unreadable files and no exact duplicates, so **0 images were removed** and all 367 were retained. Following the paper, each training source was expanded by **6 rotations (0°, 30°, 45°, 90°, 180°, 270°) × 2 flip states = 12 variants**, giving **3072 augmented training samples** from 256 sources.

2.2 Leakage safeguard — deliberate deviation from the paper

The paper augments first and splits afterwards. That ordering places rotated and flipped **copies of the same photograph** across train, validation and test, so a model is scored on images it has effectively already seen and the

result measures memorisation rather than generalisation.

Safeguard	Paper behaviour	This implementation
Split ordering	Augment, then split	Split first at source-image level , augment train only
Test set	Augmented test set	Clean, non-augmented test set
Leakage check	Not reported	Automated, PASSED , re-verified at train time

Table 2.1 — Deviation from the paper. This makes our test set strictly harder, which is central to Section 5.

Split	Source images	Samples	Augmented?
Train	256	3072	Yes (12×)
Validation	55	55	No
Test	56	56	No

Table 2.2 — Final 70/15/15 source-group split at seed 42. The leakage check asserts train source IDs are disjoint from validation and test, and that no validation or test sample is augmented.

3. Models, parameters and training configuration

Model (our label)	torchvision arch	Trainable parameters	Total params
VGG16	vgg16	134.29 M	134 290 000
ResNet18	resnet18	11.18 M	11 180 000
ResNet34	resnet34	21.29 M	21 290 000
D_ResNet	resnet50	23.52 M	23 520 000

Table 3.1 — Architectures and parameter counts, verified at runtime by the model-construction step in notebook 02. Only the final classifier is replaced with a 6-output linear layer; no architecture is hand-implemented.

VGG16 is by far the largest at 134.29 M parameters, because its three fully-connected layers (25088→4096→4096→6) dominate. It is correspondingly the slowest to train, which is worth noting when comparing training budgets across models.

Disclosure on D_ResNet. The paper proposes D-ResNet, a 50-layer bottleneck residual architecture. **The original source implementation and all training hyperparameters are not publicly available.** We use torchvision ResNet50 as a practical architectural approximation. **This is not an exact reproduction**, and no result here should be read as one.

3.1 Baseline hyperparameters, stated explicitly

Parameter	Value	Basis
Epochs	60 per model (240 across 4 models)	Paper specifies epochs 1–60
Batch size	32	Engineering choice
Optimiser	Adam	Engineering choice
Learning rate	1e-4, fixed, no scheduler	Engineering choice
Weight decay	0.0	Engineering choice — unregularised baseline
Label smoothing	0.0	Engineering choice
Loss	CrossEntropyLoss	Engineering choice
Seed	42 (single seed)	Reproducibility
Image size	224 × 224 × 3 RGB	Pretrained-weight requirement
Normalisation	ImageNet mean/std	Pretrained-weight requirement
Mixed precision	AMP in training, FP32 forced in evaluation	VGG16 produced NaN under AMP eval
Checkpoint rule	Best validation accuracy	Never selected on test

Table 3.2 — Baseline configuration. Note the baseline is a completely unregularised model, which is what motivated the tuning study in Section 4.

3.2 Epoch accounting per model

Model	Epochs run	Test used to select?	Best epoch	Best val acc	Test loss	Test acc
ResNet18	60 / 60	No — validation only	30	0.8727	1.0388	0.8036
VGG16	60 / 60	No — validation only	4	0.8182	2.7086	0.7679
ResNet34	60 / 60	No — validation only	2	0.8364	1.0040	0.7321
D_ResNet	60 / 60	No — validation only	8	0.8364	1.5089	0.6964

Table 3.3 — All four models completed the full 60 epochs and the selected checkpoint was always chosen on validation, never on test. Training accuracy reached ~1.000 by epoch 2 (VGG16 by epoch 6), so the useful training finished far earlier — this is why the tuning trials used 20 epochs.

4. Hyperparameter tuning: what was tuned and the result

4.1 Which hyperparameters were tuned

The baseline uses **zero weight decay and zero label smoothing**, on a dataset where training accuracy reaches 1.000 by epoch 2. That is exactly the regime in which regularisation should matter, so we tuned the two architecture-fair regularisation levers rather than searching a grid.

Hyperparameter	Baseline (config A)	Tuned (config B)	Fixed / not tuned	Why
Weight decay	0.0	1e-4	—	L2 penalty on weights; penalises the large weights memorisation needs
Label smoothing	0.0	0.1	—	Softens one-hot targets; discourages overconfident logits
Learning rate	1e-4	1e-4	FIXED — not tuned	Held constant to isolate regularisation
Batch size	32	32	FIXED — not tuned	Held constant
Epochs	60	20	FIXED — not tuned	Reduced from 60 to 20 on convergence evidence
Dropout	as in torchvision	as in torchvision	DELIBERATELY NOT TUNED	VGG16 already has two Dropout(0.5) layers; ResNets have none. A knob would reduce regularisation for one and add it for others, biasing the cross-architecture comparison
Optimiser	Adam	Adam	FIXED — not tuned	Held constant
Seed	42	42, 43, 44	VARIED — 3 seeds	Measures run-to-run variance
Image size	224×3	224×3	FIXED — not tuned	Pretrained-weight requirement

Table 4.1 — Exactly which hyperparameters were tuned. Two were varied (weight decay, label smoothing); the rest were deliberately held constant so that any measured change is attributable to regularisation.

4.2 Trial matrix

Dimension	Values	Count
Architectures	VGG16, ResNet18, ResNet34, D_ResNet	4
Configurations	A_baseline (wd 0.0, ls 0.0), B_reg (wd 1e-4, ls 0.1)	2
Seeds	42, 43, 44	3
Total trials	$4 \times 2 \times 3$	24
Epochs per trial	20	20
Total epochs trained	24×20	480

Table 4.2 — 24 completed trials, 480 epochs. Running config A across three seeds supplies the error bar that makes the comparison meaningful.

4.3 The test-set guard

Each trial selects its checkpoint by highest **validation** accuracy; test is read **once afterwards**. The configuration decision was made on **validation mean** across three seeds, and stands on validation. Evaluating test for all 24 trials is reporting, not leakage — choosing the maximum of 24 test scores would be leakage, and this study makes no such choice.

5. Metrics and results before hyperparameter tuning

Four metric families are reported: **accuracy**, **macro precision**, **macro recall (sensitivity)**, **macro F1**, plus **macro specificity** derived from the confusion matrix as one-vs-rest $TN/(TN+FP)$. Model selection never used the test set.

Model	Accuracy	Macro precision	Macro recall	Macro F1	Macro specificity	95% CI
ResNet18	0.8036 (45/56)	0.8093	0.8019	0.8018	0.9608	0.682–0.887
VGG16	0.7679 (43/56)	0.7725	0.7667	0.7667	0.9536	0.642–0.859
ResNet34	0.7321 (41/56)	0.7314	0.7333	0.7303	0.9464	0.604–0.830
D_ResNet (=ResNet50 approx.)	0.6964 (39/56)	0.7043	0.6944	0.6964	0.9393	0.567–0.801

Table 5.1 — BASELINE metrics before tuning. Single seed (42), 60 epochs, unregularised.

Model	Paper accuracy	Our accuracy	Gap
VGG16	0.83	0.7679	-0.0621
ResNet18	0.78	0.8036	+0.0236
ResNet34	0.83	0.7321	-0.0979
D_ResNet (=ResNet50 approx.)	0.89	0.6964	-0.1936

Table 5.2 — Paper versus our baseline. NOT like-for-like; Section 6 explains why our test set is harder.

We do not exceed the paper for any model, and we do not claim an architecture ranking: the best and weakest differ by 6 test images, their 95% Wilson intervals overlap heavily, and one test image is worth 1.8 percentage points.

6. Metrics and results after hyperparameter tuning

24 completed trials. Accuracy is reported as mean $\pm$ standard deviation across three seeds, alongside the validation mean on which the decision was based.

6.1 Accuracy after tuning

Model	A: test accuracy	B: test accuracy	Δ test	A: val acc	B: val acc
ResNet18	0.7679 $\pm$ 0.0179	0.8036 $\pm$ 0.0357	+0.0357	0.8788	0.8727
VGG16	0.7143 $\pm$ 0.0179	0.7321 $\pm$ 0.0179	+0.0179	0.8485	0.8485
ResNet34	0.7321 $\pm$ 0.0472	0.7619 $\pm$ 0.0273	+0.0298	0.8182	0.8485
D_ResNet (=ResNet50 approx.)	0.7143 $\pm$ 0.0309	0.7857 $\pm$ 0.0309	+0.0714	0.8364	0.8424

Table 6.1 — POST-TUNING accuracy, mean $\pm$ sd over 3 seeds. Every model improved; see 6.4 for whether that improvement is real.

6.2 Precision and recall after tuning — not recorded

Notebook 04 recorded **test accuracy only** for each trial. The per-trial CSV has an accuracy column but **no precision, recall or F1 columns**, and no checkpoints were saved. Precision and recall for the tuned models therefore **cannot be reconstructed from the committed artefacts**, and no estimate is offered here.

Model	Precision (baseline)	Recall (baseline)	F1 (baseline)	Precision (tuned)	Recall (tuned)
ResNet18	0.8093	0.8019	0.8018	not recorded	not recorded
VGG16	0.7725	0.7667	0.7667	not recorded	not recorded
ResNet34	0.7314	0.7333	0.7303	not recorded	not recorded
D_ResNet (=ResNet50 approx.)	0.7043	0.6944	0.6964	not recorded	not recorded

Table 6.2 — Highlighted columns are unavailable. Baseline precision and recall are measured values; tuned precision and recall were never stored. Supplying numbers here would be fabrication.

This is a limitation of the tuning experiment's instrumentation, not a claim about the models. Two properties of tuned accuracy nevertheless give a partial, defensible substitute:

- **On a near-balanced test set, macro recall equals accuracy.** The 56-image test split is distributed across six classes with no dominant class, so macro-averaged recall and accuracy agree closely in Table 5.1. Post-tuning accuracy in Table 6.1 is therefore a reasonable proxy for *macro recall* across the three seeds.
- **Precision and recall trade off against each other.** Under label smoothing the model becomes systematically less overconfident, which typically lowers precision and raises recall at fixed accuracy. So tuned precision is expected to sit slightly below tuned recall — an expectation, not a measurement.

To obtain the real values, notebook 04 would need re-running with per-class reports written to CSV alongside each trial's accuracy, and checkpoints retained so the selected weights can be re-evaluated. That is a 480-epoch re-run. The change needed is small (three lines in the trial loop); the cost is not.

6.3 All 24 trial accuracies

Model	Config	seed 42	seed 43	seed 44	mean best ep
ResNet18	A (wd 0.0)	0.7500	0.7679	0.7857	7.0
ResNet18	B (wd 1e-4, ls 0.1)	0.8393	0.7679	0.8036	10.0
VGG16	A (wd 0.0)	0.6964	0.7143	0.7321	7.0
VGG16	B (wd 1e-4, ls 0.1)	0.7321	0.7143	0.7500	7.0
ResNet34	A (wd 0.0)	0.6786	0.7679	0.7500	11.0
ResNet34	B (wd 1e-4, ls 0.1)	0.7857	0.7679	0.7321	7.0

Model	Config	seed 42	seed 43	seed 44	mean best ep
D_ResNet (=ResNet50 approx.)	A (wd 0.0)	0.6964	0.7500	0.6964	9.0
D_ResNet (=ResNet50 approx.)	B (wd 1e-4, ls 0.1)	0.8036	0.7500	0.8036	2.0

Table 6.3 — Individual trial accuracies, showing how far one architecture moves between seeds.

6.4 Did accuracy really increase?

Configuration	Mean validation (12 trials)	Mean test accuracy (12 trials)
A: wd 0.0, ls 0.0	0.8455	0.7321 ± 0.0349
B: wd 1e-4, ls 0.1	0.8530	0.7708 ± 0.0372
Δ	+0.0075	+0.0387

Table 6.4 — Configuration-level aggregates. Test rose roughly five times more than validation.

Model	A val mean	A sd	B val mean	Δ val	Verdict
D_ResNet	0.8364	0.0182	0.8424	+0.0061	inside spread — INCONCLUSIVE
ResNet18	0.8788	0.0278	0.8727	-0.0061	inside spread — INCONCLUSIVE
ResNet34	0.8182	0.0315	0.8485	+0.0303	inside spread — INCONCLUSIVE
VGG16	0.8485	0.0105	0.8485	+0.0000	inside spread — INCONCLUSIVE

Table 6.5 — Decisive test on the VALIDATION metric. Inconclusive for every architecture.

Conclusion: regularisation is not demonstrated to have helped. The direction is consistent across all four models, which argues for a small real effect. But the gain sits inside the seed spread, and validation — computed from the same models — moved only +0.0075 while test moved +0.0387. A five-to-one disagreement between the two signals points to seed variance rather than improved generalisation, so we report a null result.

6.5 What the study corrected in the baseline

Model	seed 42 (Table 5.1)	mean of 3 seeds	sd	movement (test images)
D_ResNet	0.6964	0.7143	0.0309	1.7
ResNet18	0.8036	0.7679	0.0179	1.0
ResNet34	0.7321	0.7321	0.0472	2.6
VGG16	0.7679	0.7143	0.0179	1.0

Table 6.6 — Both leading baseline models sat above their own three-seed means, so the baseline ranking is less solid than a single run suggested.

Repeating the *identical* configuration across three seeds produced test-accuracy standard deviations of 0.018 to 0.047, or 1 to 2.6 test images of movement. Any architecture gap smaller than that was never resolvable from a single seed. This is the most useful outcome of the tuning study: it replaced an unsupported ranking with a quantified uncertainty.

6.6 Limitations

- **Aggregate accuracy only.** Per-class metrics and checkpoints were not saved, so tuned models cannot be broken down by class (see 6.2).
- **One split, three seeds.** Measures training variability on a single fixed split, not split variability — likely the larger of the two at 256 unique images.
- **20 epochs versus 60.** Defensible because best epochs were early, but the baseline column is not a re-trained 20-epoch control.
- **Two configurations, not a grid.** Deliberate: on 56 test images a wider search selects noise rather than signal.

- **Carried confound.** The background/class correlation applies unchanged to every tuned model, so all figures remain upper bounds.

7. Why our accuracy is lower than the paper's

Factor	Effect on the comparison
Our test set is harder	The paper augments before splitting, so rotated and flipped copies of training photographs appear in their test set. Ours contains only clean, never-augmented, never-seen images. A meaningful part of the gap is their more permissive measurement, not our models being worse.
D_ResNet is an approximation	Original implementation unavailable, so ResNet50 stands in. The 0.19 gap on that row cannot be attributed to architecture quality . On the three architectures we can compare faithfully (VGG16, ResNet18, ResNet34) our gaps are 0.06, 0.02 and 0.10.
56-image test set	One image is 1.8 points, so 0.80 versus 0.89 is about five images. Reported accuracies at this scale carry wide intervals, ours included.
Hyperparameters unavailable	Their optimiser, learning rate, batch size and regularisation are unpublished. A different schedule can move accuracy several points at this dataset size.
Small validation set	55 validation images is enough to checkpoint but not to select hyperparameters reliably. The tuning study proves this: validation moved 0.0075 while test moved 0.0387.
Background confound	Inflates both studies equally, so it cannot explain the gap — but it means both figures are upper bounds on real mineral identification.

Table 7.1 — Six factors, roughly in order of importance. A residual gap probably remains, but isolating it requires the paper's code and settings.

Statement	Supported?
Our best baseline accuracy is 0.8036 (ResNet18, seed 42, 60 epochs)	Yes — measured, with a Wilson CI
Regularisation improved our models beyond seed noise	No — inconclusive
Our architectures are worse than the paper's	Not established — different protocol
We exactly reproduced D-ResNet	No — approximation, explicitly disclaimed
The paper's results are wrong	Not established — different protocol
The architecture ranking is resolvable here	No — intervals overlap, seed spread 1–2.6 images

Table 7.2 — Defensible claims.

8. Conclusion

The project reproduced the experimental setting of a published BSM classification study across four architectures and produced two defensible results.

- **A baseline of 0.8036 test accuracy** (ResNet18, 60 epochs, seed 42), with macro precision 0.8093, macro recall 0.8019 and macro F1 0.8018 — noting this figure sits at the top of its own seed distribution, where the three-seed mean is 0.7679 ± 0.0179 .
- **Tuned accuracy of 0.7708 ± 0.0372** as a 12-trial mean, up from 0.7321 ± 0.0349 , with regularisation parameters set to weight decay $1e-4$ and label smoothing 0.1. The gain is **inside the seed spread** and is not claimed as real.
- **A measured noise floor** of 0.018–0.047 across seeds, establishing that the architecture ranking in this dataset is not resolvable.

With 367 images and 56 test images, measurement precision — not model capacity — is the binding constraint. The most valuable product of this work is an honest account of how much of the headline accuracy can be trusted, including the parts that did not survive scrutiny.

Priority next steps: quantify the background shortcut on background-masked images; separate training variance from split variance using repeated group splits across ten or more seeds; and acquire an independent labelled test set, which is what would actually narrow the confidence intervals.

All figures generated from committed artefacts: results/metrics/final_results.csv, results/metrics/paper_reference_values.csv, results/tuning/trials.csv, results/tuning/per_config_summary.csv, results/tuning_vs_baseline/baseline_vs_tuned.csv, results/dataset_summary/dataset_summary.csv. No value in this document was transcribed by hand.